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Applied Analytics Project - Week 11 Assignment

Identification of Important Features

In our final model, the top ten predictors are listed below, in order of importance:

1. AdjEM_diff (0.115873)
2. Seed_diff (0.044444)
3. Pre-Tournament.AdjEM_diff (0.036508)
4. Pre-Tournament.RankAdjEM_diff (0.014286)
5. teamFgPct (0.014286)
6. Pre-Tournament.RankDE_diff (0.011111)
7. ftPctDiff (0.011111)
8. RankDef3PtFG (0.009524)
9. eFgPctDiff (0.009524)
10. Adjusted Defensive Efficiency Rank_diff (0.004762)

From these results, we can see that AdjEM_diff is the most important feature by a large margin. This makes sense because adjusted efficiency margin is often a strong indicator of team strength. Seed_diff is also an important feature, which makes sense because tournament seeding summarizes a team's quality based off of in-season performance. The inclusion of pre-tournament statistics in this list shows how in-season power metrics matter even when including current matchup features. Lastly, teamFgPct, eFgPctDiff, and ftPctDiff show how important shooting efficiency is in matchups.

Five Random Predictions and How the Model Generated Them

The below list shows five random predictions

	row_index	true_label	predicted_label	predicted_probability
0	1280	1	1	0.629230
1	1265	1	1	0.841512
2	1244	1	1	0.925156
3	1256	1	1	0.579948
4	1323	1	1	0.855946

In Prediction 1 (row 1280), AdjEM_diff has the largest effect. Increasing the value by 1 increased the predicted probability by 0.2288. This prediction did not have a single-feature flip, indicating that the prediction is confident and would require a more coordinated change across multiple variables to change the outcome. In Prediction 2 (row 1265), AdjEM_diff also has the largest effect. Decreasing the value by 1 increased the predicted probability by 0.059389. This prediction also did not have a single-feature flip, indicating a large change in the model would be needed to change outcome. In Prediction 3 (row 1244), most variables did not have an important effect on the model. The only variable from the list that had some impact was ftPctDiff, but the effect was small. We found that decreasing AdjEM_diff by 10 for this prediction changed the predicted probability to 0.4413, which changes the prediction outcome from 1 (a team 1 win) to 0 (a team 1 loss). In Prediction 4 (row 1256), AdjEM_diff again had the largest effect. By increasing the value by 1, predicted probability increased by 0.1290. For this prediction, decreasing AdjEM_diff by 2, decreased the predicted probability to 0.4374, changing the prediction from a 1 to a 0. Lastly, in Prediction 5 (row 1323), AdjEM_diff was again the most important feature. In this prediction, we found that decreasing AdjEM_diff by 5 changed the predicted probability to 0.4360, changing the prediction class from 1 to 0.

Protected Categories

Since this dataset only addresses March Madness matchups and not individuals, there are no protected categories like race, sex, age, religion, etc.

Bias

Bias is calculated in our dataset using the absolute value of Seed_diff. The groups are:

- Very Close (0-2)
- Close (3-5)
- Moderate (6-10)
- Large (11-20)

Below shows the bias metrics of these groups:

	group	n	accuracy	precision	recall	predicted_positive_rate	actual_positive_rate	difference_pred_minus_actual
2	Moderate (6-10)	38	0.7632	0.7778	0.9655	0.9474	0.7632	0.1842
1	Close (3-5)	37	0.5676	0.6129	0.8261	0.8378	0.6216	0.2162
0	Very Close (0-2)	29	0.6552	0.5556	0.8333	0.6207	0.4138	0.2069
3	Large (11-20)	24	0.8750	0.8750	1.0000	1.0000	0.8750	0.1250

This shows that the model predicts a team 1 win very often, with 100% of the large group predictions being class 1. For the very close group, the predicted positive rate was 0.6207, whereas the actual positive rate was only 0.4138. This difference between the minimum predicted positive rate and the maximum predicted positive rate is 0.3793. The disparate impact ratio (min/max) is 0.6207. The model is more confident in games with larger seed gaps. This is expected, but the model seems to predict a team 1 win, even in close games. This suggests the model may rely on seed too much. This demonstrates group performance bias.

Bias Removal Strategies and Retraining After Bias Mitigation

We mitigated bias by increasing weight on close games in model training. Games with an absolute value seed difference less than 2 receive weight 2.0, with all others receiving weight 1.0. This forces the model to pay more attention to close games. The model performance in the training set after this technique is below:

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Week10 Final vs Bias-Mitigation Retrain Comparison

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	model	dataset	accuracy	precision	recall
0	Week10 Final	train	0.9165	0.8944	0.9936
1	Week10 Final	validation	0.8095	0.7800	0.9750
2	Week10 Final	test	0.7031	0.7156	0.9176
3	Bias Mitigation Retrain	train	0.9165	0.8944	0.9936
4	Bias Mitigation Retrain	validation	0.7778	0.7708	0.9250
5	Bias Mitigation Retrain	test	0.7109	0.7143	0.9412

Below shows the bias metrics after mitigation for the test set:

	group	n	accuracy	precision	recall	predicted_positive_rate	actual_positive_rate	difference_pred_minus_actual
2	Moderate (6-10)	38	0.7895	0.7838	1.0000	0.9737	0.7632	0.2105
1	Close (3-5)	37	0.5946	0.6333	0.8261	0.8108	0.6216	0.1892
0	Very Close (0-2)	29	0.6207	0.5238	0.9167	0.7241	0.4138	0.3103
3	Large (11-20)	24	0.8750	0.8750	1.0000	1.0000	0.8750	0.1250

This technique slightly improved overall test accuracy and recall, but did not reduce the group imbalance issue. The predicted positives for the very close group actually increased after bias mitigation. This indicates that our bias mitigation technique did not successfully remove the group bias. Some better techniques might be to retrain without Seed_diff to reduce reliance on seed-based grouping or adjust the classification threshold to address the excessive predicted positive rate.

Based on the retraining after bias mitigation, the week 10 model had a better validation performance, but the model after bias mitigation had slightly better test accuracy and higher test recall. However, the bias between the seed groups did not improve.

Other Risks of Using the Model

There are a few risks of using this model for stakeholders. First, there is an overconfidence risk, since the model predicts class 1 very frequently. This is especially true for large seed-gap games and may lead users to believe the model is more certain than it is. There is also the risk of automation bias. This occurs when users become too reliant on the outcome and do not question it. There is also a risk that the model could become too reliant on seed differentials. It could fall into patterned predictions based on seed, rather than making new insights. There is a fairness risk. The model could systematically undervalue underdogs and smaller programs, emphasizing resource imbalances. Lastly there is a risk of misuse of this model. If the model is used for betting, the error costs are much larger than in an experimental setting, like this course.